DA233X · EXPERIMENT 2 · MODEL-BASED METHODS

Training Width in LISTA and ALISTA

Does concentrating the training-sparsity distribution buy in-distribution quality at the cost of out-of-distribution robustness? Two runs, 1,140 evaluation rows, ten seeds each — and one of the two turns out not to be a clean width experiment at all. A third run doubles the training set to see whether more data changes the answer.

runs/exp2_real + exp2_try2 + exp2_try2_moredata LISTA · ALISTA only 2026-09-07

DATA COMPLETENESS

1338 / 1338

Every model-based cell present in all three runs. The 2× run covers 3 seeds, not 10.

THE TRADEOFF, DECONFOUNDED

−4.5 dB

LISTA's OOD gain from widening at (210,300) — about 6× the gain at (25,50).

DOUBLING THE TRAINING SET

1.98×

Worse far-OOD loss for narrow-trained LISTA. Everything else barely moves.

CORRECTIONS TO THE RECORD

3

A disproved data-quality flag, a stale summary trap, and a misread figure.

01 Scope

What this covers, and what is deliberately missing from it.

Both runs contain four methods. The two model-free ones — `l2o-dm` and `l2o-rnnprop` — are excluded here because their evaluations loaded checkpoints written through the save/load collision found on 2026-09-06, so they scored optimizers that were mostly still randomly initialised. `Model_Base_L2O` uses `tf.train.Checkpoint` and never touches that path, so every LISTA and ALISTA number below is unaffected.

The consequence for the research question is worth stating plainly: §7.2 asks whether *model-free and model-based L2O differ* in how they trade in-distribution quality against out-of-distribution robustness. This analysis answers the model-based half. The comparison itself stays open until the model-free side is retrained.

Sections 03–06 cover the two main runs. §07 adds `runs/exp2_try2_moredata`, a follow-up that doubles the training set; it is model-based only, so nothing above applies to it.

All numbers are read directly from each run's `results.jsonl`, aggregated across ten seeds (three for the 2× run). Statistics are **paired by seed** wherever two conditions are compared: within a seed, all three widths are evaluated on the identical test file from the identical x_0, so

seed-to-seed problem variation cancels. With $n = 10$, $|t| > 2.26$ is significance at $p < 0.05$ two-sided.

02 Three corrections to the record

Two are traps that would put wrong numbers in the thesis. One frees a set of numbers that were flagged as suspect.

CLEARED — THE FISTA DATA-QUALITY FLAG IS WRONG

Both runs' `config.json` carry a reconstructed `num_fista_iters: 2000` and a DATA-QUALITY FLAG warning that every x^*-dependent metric may rest on an under-converged reference solve. **That reconstruction is incorrect.** The x^* cache is content-addressed on a SHA-256 of `(A, b, λ, num_iters, gap_tol)`; recomputing those keys shows every cached file matches `num_iters = 50 000, gap_tol = 1e−10` — the converged adaptive-restart solver. File counts confirm no second parameterisation exists (80 files = 10 seeds × 8 sparsities for `exp2_real` ; 110 = 10 × 11 for `exp2_try2`), and every file's mtime falls inside its own run window.

It is moot regardless. Reproducing the old plain 2000-iteration solver and differencing against the converged one gives a relative bias in f^* of 6e−18 at $p = 0.02$ and **2.1e−8 at $p = 0.40$** — four to five orders of magnitude below the effects reported here. The suboptimality gap, modified relative loss and NMSE-vs-LASSO-optimal numbers all stand as recorded.

LIVE TRAP — BOTH SUMMARY.JSON FILES ARE STALE

`runs/exp2_real/summary.json` (written 2026-08-20) and `runs/exp2_try2/summary.json` (2026-09-01) both predate the 2026-09-06 pruning of the invalidated model-free rows, and still contain 96 and 99 entries covering `l2o-dm` / `l2o-rnnprop` . `generate_result_tables.py` reads `summary.json` with a hardcoded four-method list. **Regenerating the results tables today would silently emit invalidated model-free columns.** Not fixed here — flagged for your call.

MISREAD FIGURE — THE WIDTH SPREADS WERE EQUAL; THE FRAMES WERE NOT

The 2026-09-02 write-up records that at (25,50) “ALISTA's three width curves are nearly indistinguishable” while LISTA's are genuinely width-dependent. Those figures plot R_f on a **log** y-axis, where visual separation is the log-*ratio* between curves — and measured that way the two methods are all but identical: mean width spread **LISTA 0.0614 decades vs ALISTA 0.0602 decades**.

What differed was the frame. Each method got its own figure with an independently autoscaled axis: LISTA's plotted range spans 0.20 decades, ALISTA's 1.81. So the same separation fills **30.4% of LISTA's frame and 3.3% of ALISTA's** — a 9× difference in apparent separation that is entirely autoscale. Neither method is meaningfully more width-dependent than the other on that metric at that scale.

On measures that don't depend on the frame, ALISTA is if anything the *more* width-sensitive of the two: 1.87 dB vs 0.57 dB mean spread in NMSE (already a log quantity), and 0.0133 vs 0.0034 in absolute R_f. At (210,300) the same comparison gives ALISTA 0.733 vs LISTA 0.333 decades — genuinely 2.2× more sensitive, and there the two frames are comparable (35% vs 32%), so that run's figures do not mislead.

The second half of the original reading survives intact: LISTA's R_f curve really is U-shaped and ALISTA's really is monotone, for the reason given in §06. The replacement figures in `Figs/` put both methods on one shared axis so this class of misreading can't recur.

03 Only one of the two runs is a clean width experiment

This governs how everything downstream may be read.

exp2_real

THESIS-SPEC PRIMARY RUN

Dimensions	(m,n) = (210, 300)
Noise	SNR 40 dB
Seeds	0 – 9
Narrow	p = 0.100 fixed → centre 0.1000
Medium	p ~ U(0.075, 0.20) → centre 0.1375
Wide	p ~ U(0.05, 0.30) → centre 0.1750
Rows	480 (2 × 3 × 8 × 10)

Width is confounded with centre. The three training distributions do not merely differ in spread — their means sit 0.075 apart. "Wide does better at dense *p*" is partly just "wide was trained nearer to dense *p*."

exp2_try2

PRIMER-SCALE SUPPLEMENT

Dimensions	(m,n) = (25, 50)
Noise	noiseless
Seeds	21 – 30
Narrow	p = 0.175 fixed → centre 0.1750
Medium	p ~ U(0.1125, 0.2375) → centre 0.1750
Wide	p ~ U(0.05, 0.30) → centre 0.1750
Rows	660 (2 × 3 × 11 × 10)

A clean width design. All three widths share centre 0.175, so spread is the only variable. The 2026-08-31 re-centering fixed exactly the confound above — which makes the smaller, noiseless run the methodologically stronger one for this question.

Because the runs differ in dimension, noise, training centre and test grid, nothing below compares them on an absolute number. They are compared only on *shape*: degradation measured against each width's own training centre.

04 How much of the error is the problem, not the optimizer

NMSE of the exact LASSO minimiser x^* against the true signal. Method- and width-independent — it is the floor no unrolled network can be blamed for.

Recoverability floor — NMSE(x^*, x_{true}) in dB. Near 0 dB the sparsity level is not recoverable at all.

Test p	0.02	0.05	0.10	0.1125	0.15	0.175	0.20	0.2375	0.30	0.35	0.40
exp2_real (210,300)	−44.57	−42.93	−40.07	–	−37.53	–	−34.96	–	−27.93	−21.94	−15.78
exp2_try2 (25,50)	−40.45	−43.32	−42.10	−40.48	−34.52	−30.39	−26.00	−17.50	−9.61	−6.31	−5.18

This caveat protects every dense-end number in the report. At (25,50), $p \geq 0.30$ is essentially unrecoverable — the exact minimiser itself only reaches −9.6, −6.3 and −5.2 dB — so `exp2_try2` 's dense-end "degradation" is dominated by problem difficulty rather than by any method failing. At (210,300) the floor is far more benign, reaching only −15.8 dB at $p = 0.40$. Read raw NMSE against these rows, never in isolation.

Excess over the floor

NMSE excess — dB above the exact LASSO minimiser's own recovery error. **0** = matches x^*; **negative** = recovers x_{true} *better* than x^* does.

Width	0.02	0.05	0.10	0.1125	0.15	0.175	0.20	0.2375	0.30	0.35	0.40
EXP2_REAL (210,300), 40 DB — LISTA											
narrow	7.07	5.59	3.91	–	6.31	–	13.41	–	15.62	11.59	6.75
medium	7.71	6.23	4.30	–	3.15	–	5.13	–	12.35	9.65	5.54
wide	8.61	7.17	5.31	–	4.09	–	3.47	–	8.13	7.26	4.21
EXP2_REAL (210,300), 40 DB — ALISTA											

Width	0.02	0.05	0.10	0.1125	0.15	0.175	0.20	0.2375	0.30	0.35	0.40
narrow	1.35	1.42	2.83	–	10.48	–	15.06	–	15.79	11.85	7.10
medium	0.65	3.68	1.42	–	-0.42	–	2.94	–	12.56	10.31	6.34
wide	-1.84	3.07	1.64	–	-0.94	–	0.41	–	6.76	7.43	4.95
EXP2_TRY2 (25,50), NOISELESS — LISTA											
narrow	12.07	12.19	12.93	13.10	12.47	11.25	9.54	5.34	1.47	0.27	0.01
medium	11.47	11.59	12.44	12.75	12.48	11.37	9.70	5.49	1.54	0.28	0.02
wide	10.90	11.05	12.43	13.03	13.08	11.97	10.26	5.80	1.67	0.31	0.01
EXP2_TRY2 (25,50), NOISELESS — ALISTA											
narrow	-44.63	-30.23	-20.35	-16.03	6.13	8.59	9.31	7.30	3.66	2.25	1.88
medium	-56.92	-33.03	-19.57	-15.37	6.63	9.17	9.84	7.53	3.69	2.20	1.81
wide	-54.02	-32.86	-19.05	-14.21	8.04	10.57	11.01	8.09	3.83	2.14	1.71

Two things jump out. **LISTA at (25,50) never gets within 11 dB of the LASSO optimum** anywhere in the recoverable regime, and width does not move that — it sits against a quality ceiling, which bounds how much any width effect could possibly show there. And **ALISTA goes negative**: at (210,300) medium and wide recover the true signal *better* than the exact LASSO minimiser does (−0.42 and −0.94 dB at $p = 0.15$; the −1.84 at $p = 0.02$ has a seed std of 2.38 dB and is not separated from zero). §06 explains why that is coherent rather than a bug.

05 The tradeoff, deconfounded

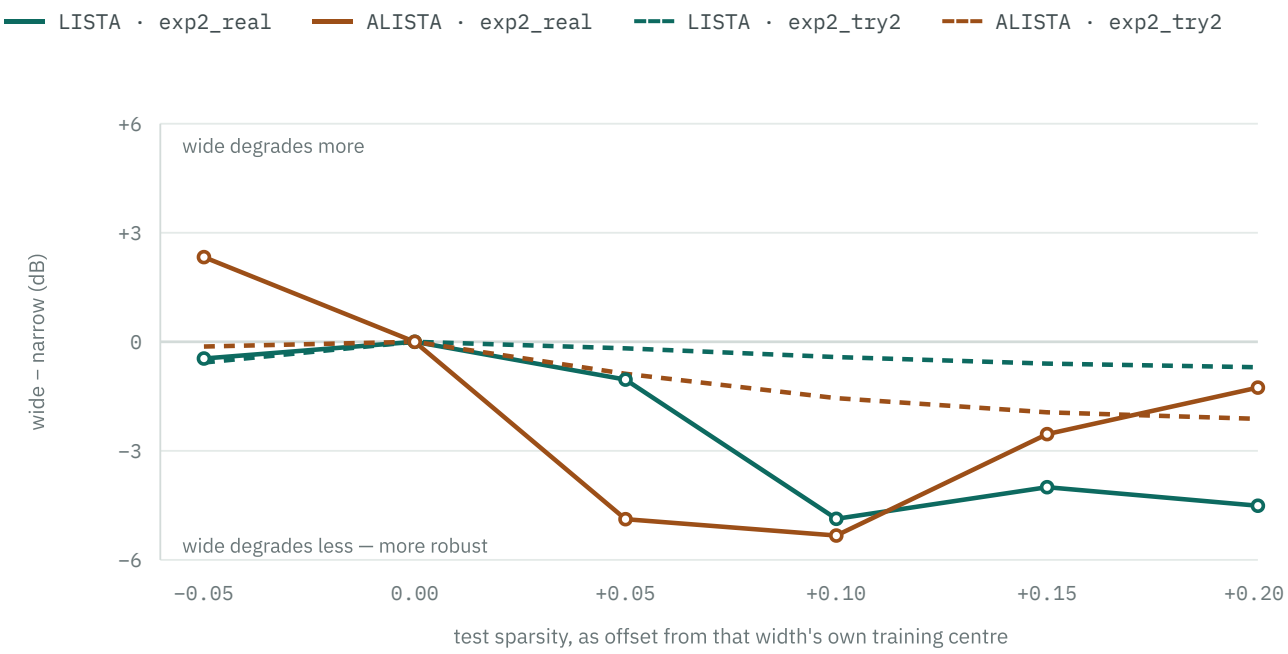
Degradation measured against each width's own training centre, on a common offset grid — so `exp2_real` 's differing centres stop doing the work.

dB of NMSE lost, relative to that width's own training centre. Per-seed linear interpolation over the test grid, then mean. Flatter = more robust. Offset 0.00 is zero by construction.

Offset from own centre	−0.05	0.00	+0.05	+0.10	+0.15	+0.20
EXP2_REAL (210,300) — LISTA						
narrow	−1.17	0.00	4.93	14.61	19.23	23.85
medium	−1.28	0.00	3.76	10.24	17.36	21.61
wide	−1.63	0.00	3.90	9.74	15.23	19.34
wide − narrow	-0.46 *	0.00	-1.04 *	-4.87 *	-4.00 *	-4.51 *
EXP2_REAL (210,300) — ALISTA						
narrow	−4.27	0.00	10.19	17.33	21.22	25.10
medium	−0.68	0.00	4.62	12.34	20.67	25.56
wide	−1.94	0.00	5.30	12.00	18.67	23.84
wide − narrow	+2.33 *	0.00	-4.88 *	-5.33 *	-2.54 *	-1.26 *
EXP2_TRY2 (25,50) — LISTA						
narrow	−6.47	0.00	5.55	9.40	12.05	13.53
medium	−6.82	0.00	5.58	9.37	11.97	13.42
wide	−7.03	0.00	5.38	8.97	11.44	12.83
wide − narrow	-0.57 *	0.00	-0.18 *	-0.42 *	-0.60 *	-0.70 *
EXP2_TRY2 (25,50) — ALISTA						
narrow	−25.34	0.00	9.43	14.14	16.79	18.12
medium	−25.32	0.00	9.19	13.68	16.21	17.48
wide	−25.47	0.00	8.55	12.59	14.85	16.00

Offset from own centre	−0.05	0.00	+0.05	+0.10	+0.15	+0.20
wide − narrow	−0.13	0.00	−0.88 *	−1.55 *	−1.94 *	−2.12 *

* = paired *t*-test across the 10 seeds, $|t| > 2.26$. Negative means wide degrades *less*. Quote these as approximate: the interpolation crosses a coarse grid — for LISTA narrow, $p = 0.20 \rightarrow 0.30$ is a ~10 dB jump and the +0.15 offset falls at its midpoint. The *sign* and rough magnitude are robust (all three of LISTA's larger offsets land at 4–5 dB); the second decimal is not.



The whole argument in one frame. Solid lines — the (210,300) 40 dB run — drop to 4–5 dB below zero: widening the training distribution genuinely buys robustness there. Dashed lines — the (25,50) noiseless run — stay within 2 dB of zero across the entire range. The benefit of widening is roughly **six times larger for LISTA at the realistic scale** than at the Primer's toy scale.

06 What the two runs say together

The full trade. ID cost is the paired wide − narrow NMSE difference at a test sparsity inside *all three* training supports; OOD benefit is the deconfounded difference at +0.20 offset. Positive ID cost = widening hurts in-distribution.

Method / run	ID cost of widening	OOD benefit at +0.20	Net
LISTA · (210,300) 40 dB	+1.40 dB *	−4.51 dB *	favourable
LISTA · (25,50) noiseless	+0.72 dB *	−0.70 dB *	a wash
ALISTA · (210,300) 40 dB	−1.19 dB *	−1.26 dB *	free — gains on both
ALISTA · (25,50) noiseless	+1.98 dB *	−2.12 dB *	a wash

ID cost is measured at $p = 0.10$ for `exp2_real` — narrow's fixed training point, and inside both medium's [0.075, 0.20] and wide's [0.05, 0.30] — and at the shared centre $p = 0.175$ for `exp2_try2`. All four are significant at 10/10 or 0/10 seeds. One asymmetry to keep in view: `exp2_try2`'s ID comparison is centre-against-centre, while `exp2_real`'s is narrow-at-its-centre against wide 0.075 off its own — the standard within-support operationalisation, but not as clean as try2's.

The defensible claim

The out-of-distribution benefit of widening the training distribution is **about six times larger for LISTA at (210,300)/40 dB than at (25,50)/noiseless** (−4.5 vs −0.7 dB). For ALISTA it runs the other way — about 1.7× larger at the *small* scale (−2.1 vs −1.3 dB). At the small noiseless scale the ID cost cancels the OOD gain almost exactly for both methods: widening is a wash. At (210,300) it is clearly worth doing — and for ALISTA it costs nothing at all.

The hypothesis this pair of runs cannot settle

It is tempting to say the tradeoff's *sign* flips with scale and noise — that widening is near-free at realistic dimensions and pointless at toy ones. The data lean that way, but `exp2_real`'s ID comparison is not centre-against-centre, and the two runs differ in dimension *and* noise *and* centre simultaneously. Settling it needs one thing: **an `exp2_real` re-run with all three widths sharing a centre** — the same fix `exp2_try2` already applies. That is the single highest-value follow-up this analysis points to.

LISTA and ALISTA disagree, consistently, and for a reason

Across every width in both runs, ALISTA is **better at recovering the signal** (NMSE 1.1–5.0 dB lower, 0/10 seeds contradicting) and **worse at minimising the LASSO objective** (R_f higher, 10/10 seeds). At (210,300) the head-to-head at each width's own centre runs −1.08, −3.57 and −5.03 dB in NMSE against +0.0050, +0.0039 and +0.0016 in R_f.

The mechanism is ALISTA's support selection. `shrink_ss` exempts the top-q% entries from shrinkage each layer, which is precisely a debiasing step relative to the LASSO minimiser: less shrinkage bias moves the estimate closer to x_{true} and further from f^*. That is why ALISTA can post *negative* excess over the floor in §04 — it is not trying to minimise the objective the floor is defined by. It also makes narrow-ALISTA's pattern at (210,300) coherent: best R_f at $p = 0.10$ (0.0135 vs 0.0153 and 0.0161) but the worst NMSE there — the narrow-trained model learned less debiasing.

One observation reported without explanation: **ALISTA-narrow is the worst of the three widths at its own training point at (210,300), but the best at (25,50).** A clean sign flip between the runs.

07 Does doubling the training set change any of this?

A follow-up run repeats the (25,50) setting with 64,000 training instances instead of 32,000. Short answer: no — except in one place, where it makes things worse.

`runs/exp2_try2_moredata` is identical to `exp2_try2` in every respect — dimensions, noise, the three widths all centred on 0.175, the same eleven test sparsities — except that `train_size` is 64,000 rather than 32,000. It covers **seeds 21–23 only**, not the full ten.

Its test files are **byte-identical** to the baseline's for those three seeds — every one of its 198 cells has a counterpart with an exactly matching x^* floor. So this is a genuinely paired comparison, cell against matching cell, not two independently drawn samples. Every number below is therefore compared against `exp2_try2` *restricted to seeds 21–23*, never against its ten-seed aggregate.

Paired R_f ratio, 64,000 ÷ 32,000 training instances — median over the 3 shared seeds. Below 1.00 means the doubled training set did better.

Method / width	0.02	0.05	0.10	0.1125	0.15	0.175	0.20	0.2375	0.30	0.35	0.40
LISTA · narrow	0.91	0.92	0.93	0.92	0.92	0.93	0.94	0.97	1.25	1.60	1.98
LISTA · medium	0.93	0.93	0.94	0.93	0.94	0.94	0.94	0.94	0.97	0.99	1.00
LISTA · wide	0.93	0.94	0.94	0.94	0.93	0.94	0.94	0.94	0.96	0.99	1.01
ALISTA · narrow	1.15	1.11	1.13	1.02	1.00	0.99	0.99	0.99	0.99	1.00	1.05
ALISTA · medium	0.99	1.00	0.99	1.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ALISTA · wide	1.05	1.04	1.06	1.05	1.00	1.01	1.00	1.00	1.00	1.00	1.00

Figure: `Figs/experiment2_moredata_effect.pdf`. Eq. 11 tells the same story — paired NMSE change (64k − 32k) is −0.32 / −0.26 / −0.28 dB for LISTA narrow/medium/wide and +0.39 / +0.32 / +0.51 dB for ALISTA, all within ±0.5 dB (`experiment2_moredata_effect_nmse_true.pdf`). Eq. 9 needs no separate figure: it is $R_f \times$ mean(f^*_q), and f^*_q depends only on the test file, so within a test sparsity the two are exactly proportional and their paired ratios are identical.

What it says

ALISTA is unmoved — 1.00 across the board wherever the measurement is meaningful. **LISTA gains a uniform 6–8%** at every test sparsity up to 0.2375, for all three widths alike. Then the widths separate sharply: at $p \geq 0.30$ medium and wide stay flat at 0.96–1.01, while **narrow degrades to 1.25, 1.60 and finally 1.98** — twice the relative loss it had with half the data.

That is the width story from §06 again, sharpened rather than softened. More data does not teach a narrow-trained LISTA to generalise; it teaches it the narrow distribution more thoroughly, and the far-OOD cost roughly doubles. The lesson for the thesis is that widening the training distribution and enlarging it are not substitutes — at this scale, only the first buys robustness, and the second amplifies whatever the first decided.

A measurement trap this run exposed

ALISTA reaches *exact recovery* on (25,50) noiseless at $p \leq 0.10$ — baseline NMSE of −61 to −97 dB. Both runs sit at numerically zero error there, so the dB difference between them is float noise, not an effect. Left in, those three columns alone swing ALISTA-narrow's mean NMSE change from ≈0 to **+4.4 dB** and stretch a shared axis until every other curve reads as flat. They are excluded from the Eq. 11 figure and its summary, at the level of whole (method, sparsity) columns rather than individual cells — at the threshold boundary only some seeds fall past it, and keeping the rest would build that point from a biased subset of seeds.

How far to trust it

Three seeds, not ten, so bands are min–max rather than IQR and no significance testing was done. And 2× is a modest factor — this bounds the effect of doubling, not of an order of magnitude. What it does establish is that the width effect in §05–06 is not simply a data-starvation artefact that more training instances would dissolve.

Provenance. All figures computed from `runs/exp2_real/results.jsonl` (480 rows), `runs/exp2_try2/results.jsonl` (660 rows) and `runs/exp2_try2_moredata/results.jsonl` (198 rows), read directly rather than through any run's `summary.json`, for the reason given in §02. Seeds 0–9, 21–30 and 21–23 respectively; every model-based cell present, none imputed.

Two records need editing and were deliberately not touched: the disproved `num_fista_iters` DATA-QUALITY FLAG in both `config.json` files, and the stale `summary.json` that

`generate_result_tables.py` still reads with a hardcoded four-method list.